All-optical fiber speckle sensing of multi-point perturbations using genetic algorithms

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Abstract Speckle-based fibre sensing enables distributed measurements by exploiting multimode interference, but conventional imaging limits acquisition speeds to the kilohertz range. We present a high-speed, all-optical interrogation method using a digital micromirror device (DMD) as a programmable spatial filter, optimized in situ via evolutionary algorithms. The speckle field is split into complementary regions and detected differentially with photodetectors, enabling signed optical weighting and direct electrical readout. This approach achieves sensitivity to fibre perturbations at megahertz rates while bypassing camera-based acquisition and digital processing, enabling real-time, high-bandwidth distributed sensing.

Sensing and Computation principle

A DMD, acts as an in-situ trainable layer. By capturing complementary regions of the modulated light in a differential scheme, we achieve signed computation capabilities.

Because perturbations at different positions x_i in an optical fibre cause different changes to its transmission matrix, for small perturbation $\delta_i = [\delta_1, \ldots, \delta_N]$, the output intensity, modulated by a DMD mask $\vec{w}$ can be expressed linearly as:

$$\boxed{\vec{I}_{out} \approx \sum_{i=1}^{N} \delta_i (\vec{w} \odot \vec{I}'_i),}$$

$\vec{I}'_i$ represents the first-order sensitivity pattern associated with perturbation δ_i

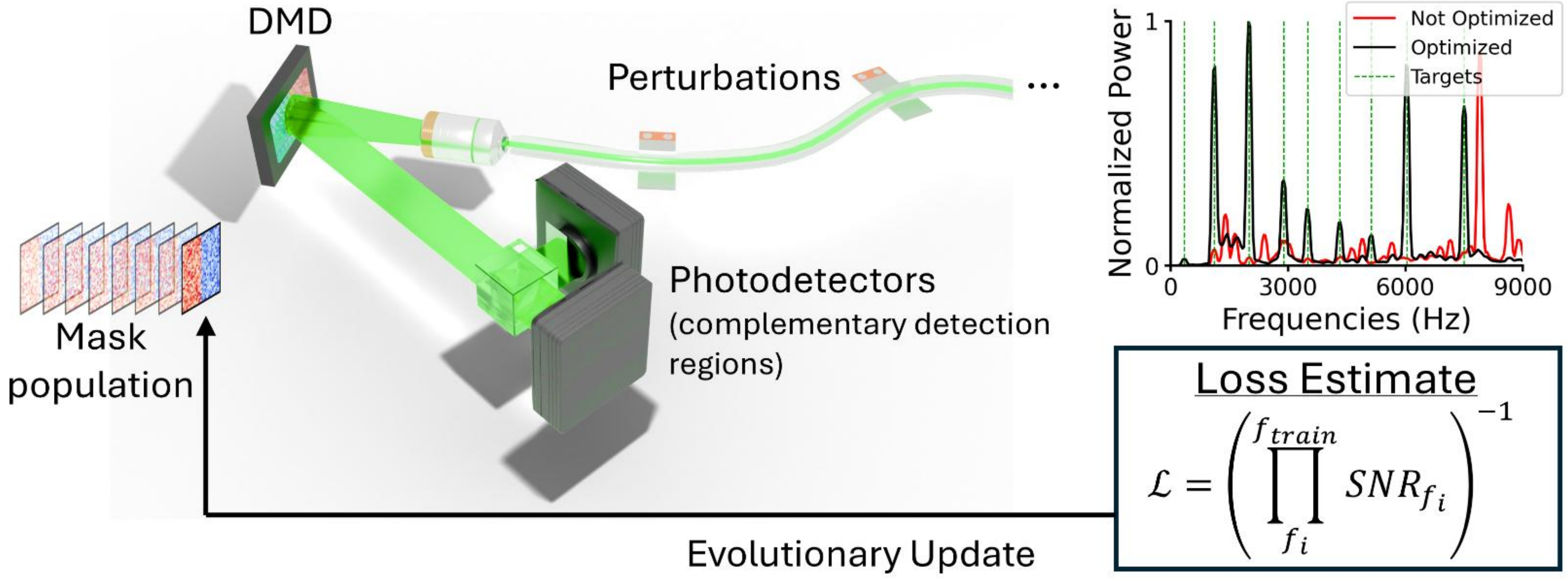


Figure 1. Schematic of the experimental setup. The output specke is directed to a DMD that modulates the output field. Light is directed towards two photodetectors, and measured using a differential measurement scheme, allowing for signed computation.

Training

We want to optimize $\vec{w}$ so that it maximizes the response to a specific point, while suppressing others. For that we use evolutionary algorithms (GA and CEM) to iteratively increase the SNR of target frequencies playing at a predefined location

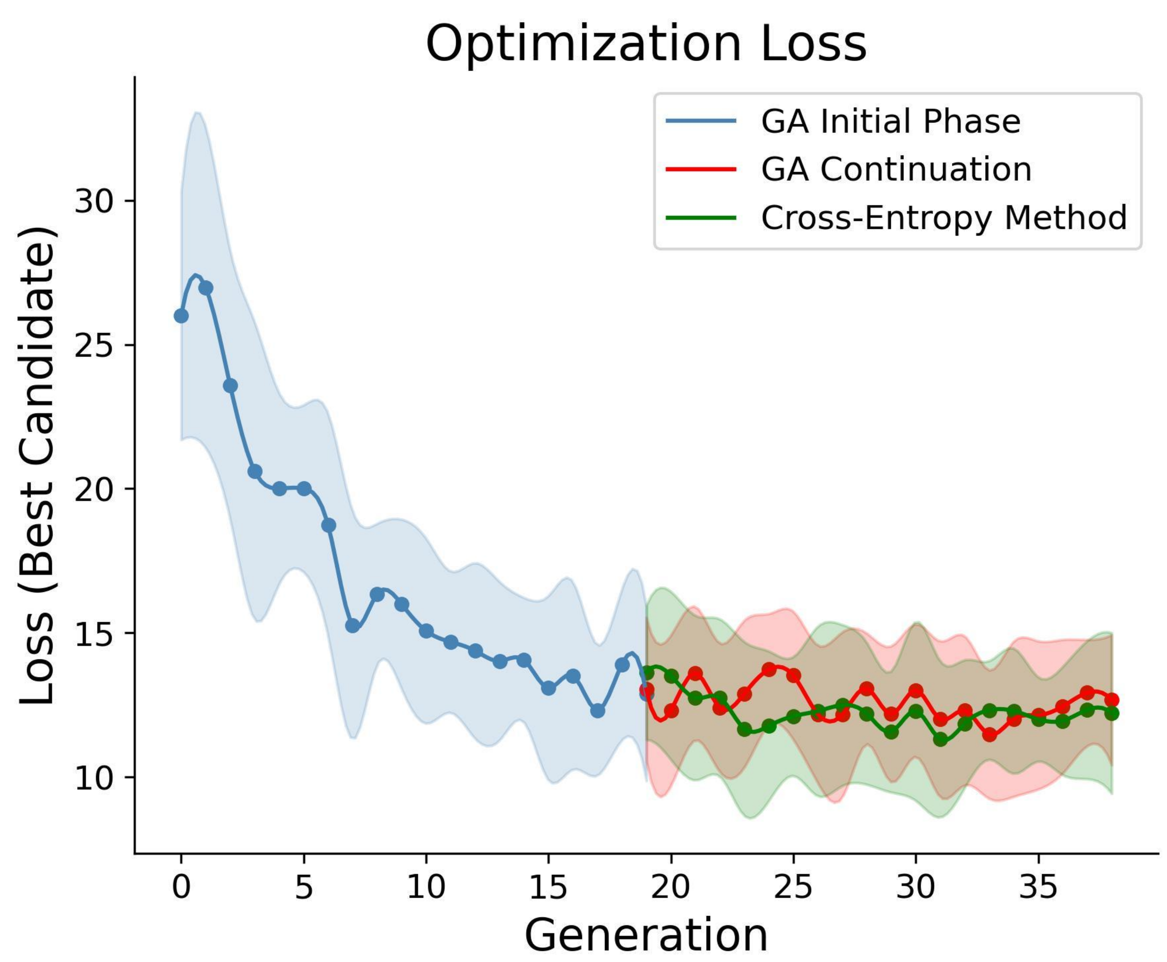
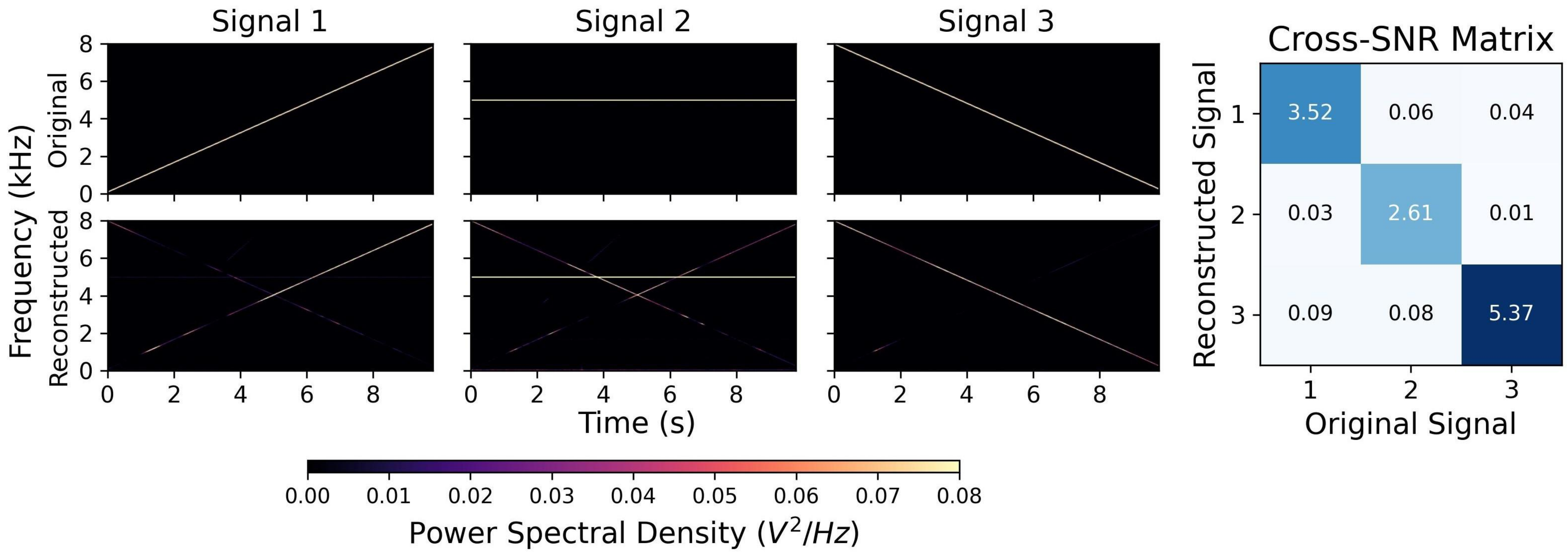


Figure 2. Loss of the best candidate over the each of training genereations, for 10 total training routines.

Results

The system isolates and reconstructs individual perturbations using in-situ optimized masks, suppressing concurrent perturbations at other positions in an all-optical scheme. Because the signal is captured using photodetectors, this platform has the potential to expand beyond the MHz regime.



	1	2	3
1	3.52	0.06	0.04
2	0.03	2.61	0.01
3	0.09	0.08	5.37

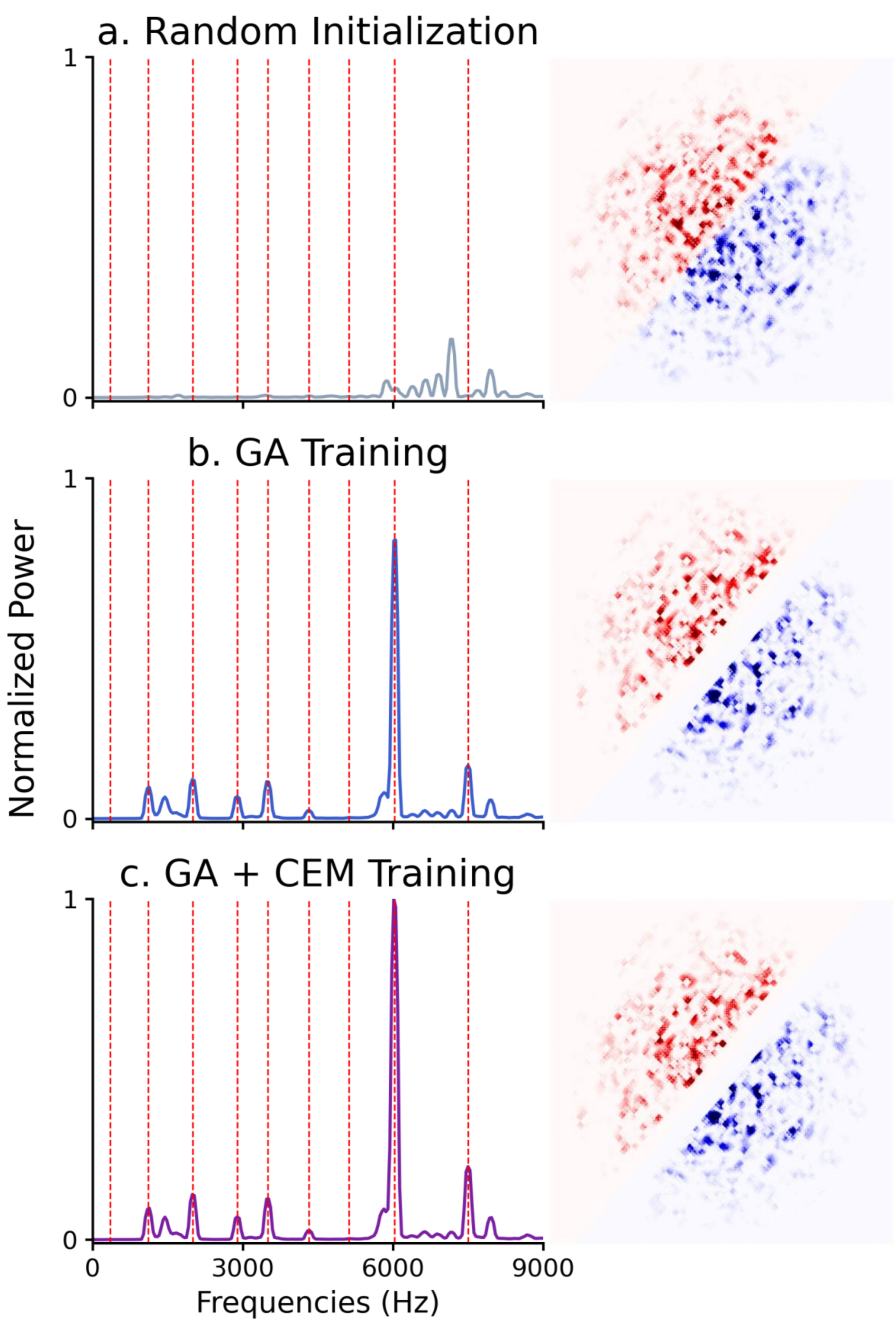


Figure 3. Reconstructed (bottom) spectrograms for simultaneous signals playing at three deformation points in the optical fibre. These signals were obtained by placing the optimized masks for each location at a time in the DMD, while the three signal (top) were playing at the same time. On the right we see how spectral power evolves with training for a specific train location.

Concluding Remarks

We demonstrate an all-optical edge-computing approach for speckle sensing that enables real-time separation and reconstruction of distributed perturbations without camera-based acquisition. By embedding computation directly in the optical domain, this framework overcomes bandwidth and latency limitations, opening new pathways for high-speed, adaptive sensing systems.

References

[1] Lopes, Tomás, et al. "Harnessing speckle optical fiber sensors through high-frequency interrogation with an event-based camera." *29th International Conference on Optical Fiber Sensors.* Vol. 13639. SPIE, 2025.

[2] Chapalo, Ivan, et al. "Advances in optical fiber speckle sensing: a comprehensive review." Photonics. Vol. 11. No. 4. MDPI, 2024.

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